FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
FULL TEST – X
PAPER –2
TEST DATE: 15-05-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A

Sol. From conservation of charge

$$-\frac{dq}{dt} = 4\pi r^2 j = 4\pi r^2 \sigma E$$

$$\Rightarrow -\frac{dq}{dt} = 4\pi r^2 \sigma\left(\frac{q}{4\pi\varepsilon_0\varepsilon_r r^2}\right) = \frac{\sigma q}{\varepsilon_0\varepsilon_r}$$

$$\Rightarrow q = q_0 e^{-\frac{\sigma}{\varepsilon_r\varepsilon_0}t}$$

$$\Rightarrow I = \frac{-dq}{dt} = \frac{\sigma q_0}{\varepsilon_r\varepsilon_0} e^{-\frac{\sigma}{\varepsilon_r\varepsilon_0}t}$$

2. C

Sol. Buoyant force decreases due to acceleration of container

3. A

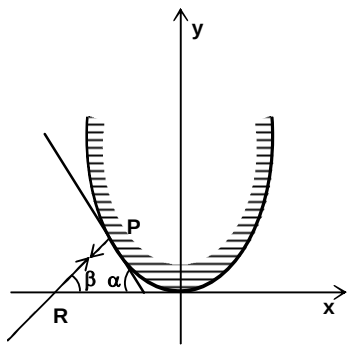
Sol. Point of intersection of two curves $y = \frac{x^2}{4}$ and y = x +3 is (–2, 1)

$\frac{dy}{dx} = \frac{x}{2}$ i.e. $\left(\frac{dy}{dx}\right)_{x=2} = \frac{-2}{2} = -1$

tan β = slope of incident ray RP = 1 or $\beta = 45°$

From geometry we can show that $\angle i = \angle r = 0°$ i.e., the ray is incident normally.

Hence, the desired unit vector : $A = \frac{1}{\sqrt{2}}(-\hat{i} - \hat{j})$



4. C

Sol. Energy photon is

$E_\gamma = \frac{12431}{0.125} = -0.09945$ MeV

Mass defect of reaction is

$\Delta m = 239.052158 - 235.04925 - 4.002603$

$\Delta m = 0.00563$ amu

Energy released is

$E = 0.00563 \times 9.31$ MeV

$\Rightarrow E = 5.24434$ MeV

Kinetic energy distributed in $^{235}_{92}U$ and $^{4}_{2}He$ is approximately

$E_k = E - E_y = 5.24434 - 0.09945 = 5.1449$ MeV

Thus kinetic energy of α-particle is

$k_\alpha = \left(\frac{A-4}{A}\right)E_k = \frac{235}{239} \times 5.1449$ MeV

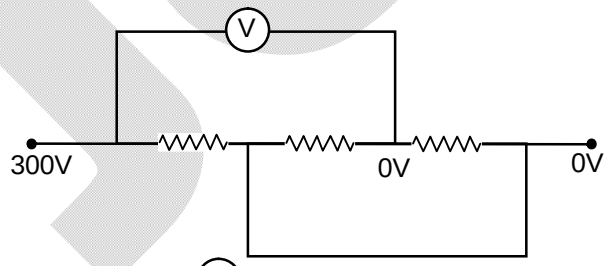
$\Rightarrow k_\alpha = 5.0588$ MeV

5. A,C, D

Sol. The system can be assumed to be combination of a uniform sphere with four small sphere of -ve mass.

6. A, B, C, D

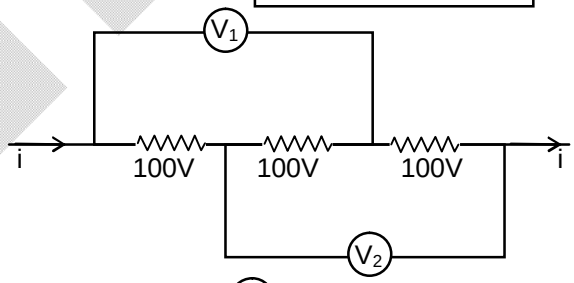
Sol. Ideal voltmeter ($R_V = \infty$)
Ideal ammeter ($R_A = 0$)
Reading 300 V



$i = \frac{300}{300} = 1A$

$V_1 = 100 + 100 = 200$ V

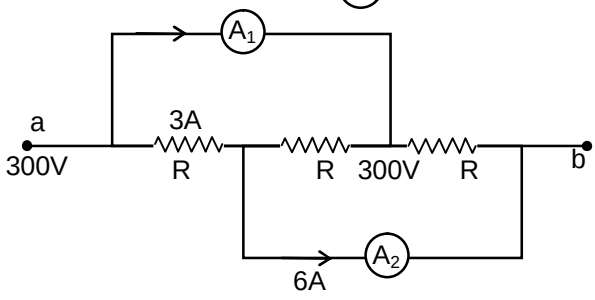
$V_2 = 100 + 100 = 200$ V



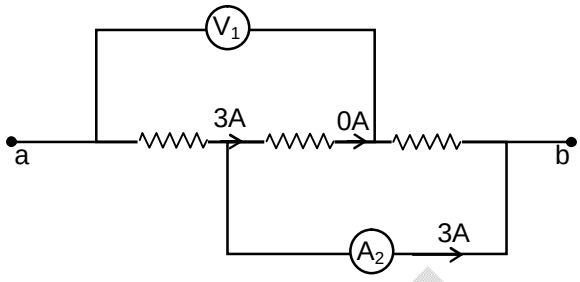
All resistance are in parallel combination

$i = \frac{300}{100/3} = 9A$

Reading of ammeter is 6A



$$i = \frac{300}{100} = 3A$$



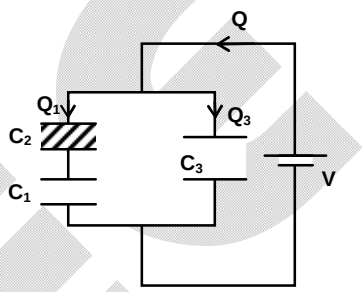
7. A, B, D

Sol. $q = C_{eq}V$

$$C_1 = \frac{A\varepsilon_0}{d}, C_2 = \frac{2A\varepsilon_0}{d} \text{ and } C_3 = \frac{A\varepsilon_0}{2d}$$

$$Q_1 = \frac{2}{3}\frac{A\varepsilon_0 V}{d} \Rightarrow \sigma_1 = \frac{Q_1}{A/2} = \frac{2Q_1}{A} = \frac{4\varepsilon_0 V}{3d}$$

$$Q_3 = \frac{A\varepsilon_0 V}{2d} \Rightarrow \sigma_2 = \frac{Q_3}{A/2} = \left(\frac{\varepsilon_0 V_0}{d}\right)$$



SECTION – B

8. 1

Sol. For sphere to just start slipping

$$\mu = \frac{\tan\theta}{1+\frac{mR^2}{I}} = \frac{2}{7}\tan\theta$$

$$a_{sphere} = \frac{g\sin\theta}{1+\frac{I}{mR^2}},$$

$$a_{cylinder} = g\sin\theta - \mu g\cos\theta = \frac{5}{7}g\sin\theta$$

Both cylinder and sphere have same acceleration

9. 2

Sol. Range = $v_0 \times$ time of flight (t)

$$\Rightarrow t = \frac{2v_0^2}{gv_0} = \frac{2v_0}{g}$$

If y is the height of balloon at any instant t and $\frac{dy}{dt}$ its velocity then

$$y = -\left(\frac{dy}{dt}\right)\left(\frac{2v_0}{g}\right) + \frac{1}{2}g\left(\frac{2v_0}{g}\right)^2$$

$$\Rightarrow \frac{dy}{dt} + \frac{g}{2v_0}y - v_0 = 0$$

$$\frac{dy}{\left(v_0 - \frac{g}{2v_0}y\right)} = dt$$

$$\Rightarrow \quad -\frac{2v_0}{g}\ell n\left(v_0-\frac{g}{2v_0}y\right)=t+c$$

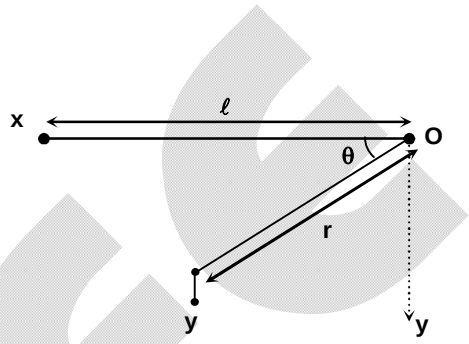
At t = 0, y = 0 $\Rightarrow$ $c=-\frac{2v_0}{g}\ell nv_0$

Simplifying, we get

$$y=\frac{2v_0^2}{g}\left[1-e^{-gt/2v_0}\right]$$

10. 9

Sol. Let the second nail be located at a distance r from the first nail and angle θ below horizontal as shown. Now speed of the bob at its lowest position can be found using conservation of energy.



$$\frac{1}{2}mv^2=mg[r\sin\theta+\ell-r]$$

$$\Rightarrow v^2=2g(r\sin\theta+\ell-r) \qquad \ldots(i)$$

The particle will compete circle about the second nail if $v^2=5g(\ell-r)$ …(ii)

From (i) and (ii)

$2(r\sin\theta+\ell-r)=5(\ell-r)$

$\Rightarrow 3r+2r\sin\theta=3\ell$ …(iii)

Now coordinates of the nail are

$x = r\cos\theta$, $y = r\sin\theta$

Using these in (iii), we get

$$3\sqrt{x^2+y^2}+2y=3\ell$$

Or $9(x^2+y^2)=(3\ell-2y)^2$

11. 9

Sol. $\omega_1 = 2\omega_2 = 4\omega_3$, if $\omega_3 = \omega$

then $\omega_1 = 4\omega$, $\omega_2 = 2\omega$

$a_1 = 16\omega^2 r$, $a_2 = 4\omega^2 2r$, $a_3 = \omega^2 3r$

$\Rightarrow$ $a_1 : a_2 : a_3 = 16 : 8 : 3$

12. 3

Sol. At t = 3

$$\text{Slope}=\frac{d\left(\frac{1}{V}\right)}{dt}=-1$$

$$\Rightarrow \quad -\frac{1}{V^2}\frac{dV}{dt}=-1$$

$$\Rightarrow \left(\frac{1}{\sqrt{3}}\right)^2\frac{dV}{dt}=1\Rightarrow a=\frac{dV}{dt}=3\ m/s^2$$

13. 3

Sol. $Q_1=\frac{n}{2}\times10^2\times2+0.5m\times2$

$500 = 100\,n + m$ …(i)

$Q_2 = \frac{n}{2} \times 10^2 \times 2 + 0.5 \times m \times 1 + m \times 80 + m \times 1 \times 1$

$16600 = 100\,n + 81.5\,m$ …(ii)

$24150 = 8050\,n,\ n = 3$

SECTION – C

14. 0.50

Sol. $v_1\rho_w g = (v - \Delta v)\rho_i g + \Delta v(\rho_m)g$

$v_2\rho_w g = (v - \Delta v)\rho_i g + \Delta v(\rho'_m)g$

$$\frac{v_1}{v_2} = \frac{(v-\Delta v)\rho_i + \Delta v\rho_M}{(v-\Delta v)\rho_i + \Delta v\rho'_M} = \frac{\left(\frac{v}{\Delta v} - 1\right)\rho_i + \rho_m}{\left(\frac{v}{\Delta v} - 1\right)\rho_i + \rho_m}$$

$$= \frac{\left(\frac{1000}{20} - 1\right) \times 0.9 + 4.9}{\left(\frac{1000}{20} - 1\right) \times 0.9 + 1.9} = \frac{49}{46}$$

15. 9.80

Sol. Mass of cavity is more in cube A than in cube B.

16. 0.10

17. 0.20

Sol. (Q.16-17)

Acceleration of cylinder is given by

$$a = \frac{mg\sin\theta\sin\alpha}{m + \frac{m}{2}} = \frac{2g\sin\theta\sin\alpha}{3} = \frac{g}{6}$$

friction along x-axis : $f_x = mg\sin\theta\cos\alpha$

friction $\perp$ to x -axis : $f_y = \frac{mg\sin\theta\sin\alpha}{3}$

$\therefore$ Friction $= \sqrt{f_x^2 + f_y^2} = \frac{\sqrt{7}mg}{6}$

Chemistry

PART – II

SECTION – A

18. A

Sol. $AgA(s) \rightleftharpoons Ag^{+}(aq) + A^{-}(aq)$

s s – x $K_{sp} = s(s-x)$

Hydrolysis of A^{-}

$A^{-}(s) + H_2O \rightleftharpoons HA + OH^{-}$

s – x x x $K_h = \dfrac{K_w}{K_a} = \dfrac{x^2}{s-x}$

$K_h = \dfrac{10^{-10}}{s - 10^{-5}} = 10^{-4}$

$s = 1.1 \times 10^{-5}$

$K_{sp} = s(s-x) = 1.1 \times 10^{-11}$

19. C

Sol. $\Delta G^{o} = \Delta H^{o} - T\Delta S^{o} = -2.303RT\log K_P$

$K_P = 10$

T = 371 K

20. D

Sol. NF_3 does not hydrolyze due to lack of vacant orbital on either F or N.

NCl_3 can undergo hydrolysis due to presence of vacant d-orbital of Cl and forms NH_3 and HOCl.

21. A

Sol. $(NH_4)_2 Cr_2O_7 \xrightarrow{\Delta} Cr_2O_3 + N_2 + H_2O$

$NH_4NO_2 \xrightarrow{\Delta} N_2 + H_2O$

22. A, C, D

Sol. $[NiCl_4]^{2-} \rightarrow sp^3\,(\text{tetrahedral})$

23. B, C, D

Sol. Neoprene is a polymer of chloroprene.

24. A, B

Sol. $-SO_3H$ and $-CN$ are metadirecting and deactivating groups.

SECTION – B

25. 2

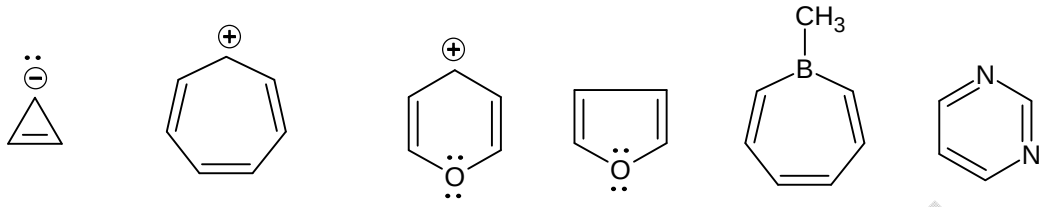
Sol. (c) and (h) cannot be prepared

26. 3

Sol. Iron Pyrites, Argentite, Cinnabar

27. 6

Sol. Aromatic compounds are



28. 15

Sol. $6HCHO + 4NH_3 \longrightarrow (CH_2)_6 N_4 + 6H_2O$

Urotropine

Number of C-atom = x = 6

Number of N-atom = y = 4

$\therefore \frac{x}{y} \times 10 = 15$

29. 5

Sol. $\Delta E = 13.6Z^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$25.7 = 13.6Z^2\left[\frac{1}{4} - \frac{1}{n^2}\right]$

$8.7 = 13.6Z^2\left[\frac{1}{a} - \frac{1}{n^2}\right]$

(1) by (2) gives

$n^2 = 25.32 \Rightarrow n = 5$

30. 133

Sol. $8MnO_4^- + 3S_2O_3^{2-} + H_2O \longrightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^-$

$-\frac{1}{8}\frac{d\left[MnO_4^-\right]}{dt} = \frac{1}{6}\frac{d\left[SO_4^{2-}\right]}{dt}$

$\frac{-\frac{d\left[MnO_4^-\right]}{dt}}{\frac{d\left[SO_4^{2-}\right]}{dt}} = \frac{8}{6} = 1.33 = x$

$100x = 100 \times 1.33 = 133$

SECTION – C

31. 10.00

32. 2.00

33. 6.00

34. 4.00

Mathematics

PART – III

SECTION – A

35. A

Sol. Let $y = \frac{x^2 + bx + 1}{x^2 + 2x + b}, (b > 1)$

$\Rightarrow (y - 1)x^2 + (2y - b)x + (by - 1) = 0$

$\Rightarrow D \geq 0$

$\Rightarrow (4 - 4b)y^2 + 4y + (b^2 - 4) \geq 0$

Since, f(x) and $\frac{1}{f(x)}$ have the same bounded set as their range

Thus, $(4 - 4b)y^2 + 4y + (b^2 - 4) \geq 0$ have roots α and $\frac{1}{\alpha}$

$\therefore$ Product of roots = 1

$\Rightarrow \frac{b^2 - 4}{4(1-b)} = 1$

$\Rightarrow b^2 - 4 = 4 - 4b$ or $b^2 + 4b - 8 = 0$

$b = \frac{-4 \pm \sqrt{16 + 32}}{2} = \frac{-4 \pm 4\sqrt{3}}{2} = 2\sqrt{3} - 2$

36. B

Sol. We rewrite function f(x) as $f(x) = \sqrt{a^2 + 1}\sin(ax + \varphi)$, where $\varphi = \arctan\frac{1}{a}$. Its least positive period is $\frac{2\pi}{a}$, and its amplitude is $\sqrt{a^2 + 1}$. By symmetry of the figure enclosed by the graphs of the functions f(x) and g(x), we can change the figure into a rectangle with length $\frac{2\pi}{a}$ and width $\sqrt{a^2 + 1}$ using the cut-and-paste method.

Therefore its area is $\frac{2\pi}{a}\sqrt{a^2 + 1}$

37. A

Sol. $\sqrt[2n+1]{f(x)\cdot\underbrace{1\cdot 1\cdot .....\cdot 1}_{2n \text{ terms}}} \leq \frac{f(x) + 1 + ... + 1}{2n + 1} \Rightarrow \int_a^b \sqrt[2n+1]{f(x)}dx < \frac{1 + 2n}{2n + 1} = 1$

So, $\left(\int_a^b \sqrt[2n+1]{f(x)}dx\right)\left(\int_a^b \sqrt[2n+3]{f(x)}dx\right)\left(\int_a^b \sqrt[2n+5]{f(x)}dx\right) < 1$

38. C

Sol. Let $t = 2^x$. Clearly, $t \in [2, 4]$, and $|t - a| < |5 - t|$ holds for all $t \in [2, 4]$. We eliminate the absolute value sign by taking squares on both sides

$|t - a| < |5 - t| \Leftrightarrow (t - a)^2 < (5 - t)^2 \Leftrightarrow (2t - a - 5)(5 - a) < 0$

For given real a, let $f(t) = (2t - a - 5)(5 - a)$

Notice that f(t) is a linear function of t (or constant 0 when a = 5)

To have $f(t) < 0$ for all $t \in [2, 4]$, it suffices to require that $\begin{cases} f(2) = (-1 - a)(5 - a) < 0 \\ f(4) = (3 - a)(5 - a) < 0 \end{cases}$ which gives

$3 < a < 5$. Therefore, then range of a is $3 < a < 5$

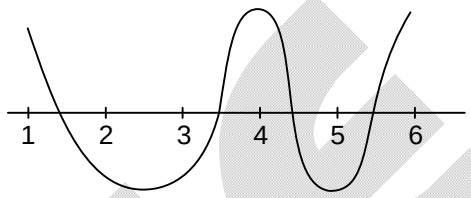
39. A, B, C

Sol. $|A + |A| \text{ Adj } A| = 0 \Rightarrow |(\text{Adj } A)A + |A|(\text{Adj } A)^2| = 0$
$\Rightarrow |1 + (\text{Adj } A)^2| = 0 \Rightarrow bc - ad + 1 = 0$ and $-d = a$
So, $\vec{u}' \cdot \vec{v}' = 1$

Now, $\left|A - |A| \text{Adj } A\right| = \frac{1}{|\text{Adj } A|}\left|\text{Adj } A \cdot A - |A|(\text{Adj } A)^2\right| = \left|I - (\text{Adj } A)^2\right| = 4$

40. A, B

Sol. $x^2y + xy^2 = 5 + x^3 + y^3$
$\Rightarrow (x - y)^2(x + y) = -5$
So, $(x - y)^2 = 1$ and $x + y = -5$
$\Rightarrow x - y = \pm 1$ and $x + y = -5$
$(x, y) = (-2, -3)$ and $(x, y) = (-3, -2)$



41. A, B, C

Sol. $3^{\alpha} = 4\lambda_1 + (-1)^{\alpha}$
$5^{\beta} = 4\lambda_2 + 1$
$7^{\gamma} = 8\lambda_3 + (-1)^{\gamma}$

SECTION – B

42. 7

Sol. $\beta = 1$, $\sin x = \frac{|x|}{7}$ total 7

43. 1

Sol. We take BK = CH on BE and join OB, OC and OK
From the property of the circumcenter of a triangle, we know that $\angle BOC = 2\angle A = 12\text{-}0°$. From the property of the orthocenter of a triangle, we get $\angle BHC = 180° - \angle A = 120°$
So $\angle BOC = \angle BHC$. Then four points B, C, H and O are concyclic
Hence $\angle OBH = \angle OCH$
In addition, OB = OC and BK = CH
Therefore, $\Delta BOK \cong \Delta COH$. It follows that $\angle BOK = \angle COH$ and OK = OH
So, $\angle OKH = \angle BOC = 120° \Rightarrow \angle OKH = \angle OHK = 30°$
In ΔOKH, by the sine rule, we get $KH = \sqrt{3}OH$. In view of BM = CN and BK = CH, we get KM = NH and $MH + NH = MH + KM = KH = \sqrt{3}OH$
Therefore, $\frac{MH + NH}{OH} = \sqrt{3}$

44. 3

Sol. $x = \sec^{4k} x + \text{cosec}^{4k} x \geq \frac{2 \cdot 2^{2k}}{\sin^{2k} \cdot 2x}$

$y = \sec^{6n} y + \text{cosec}^{4n} y \geq \frac{2 \cdot 2^{3n}}{\sin^{2n} \cdot 2y}$

$z = \sec^{8m} z + \text{cosec}^{8m} z \geq \frac{2 \cdot 2^{4m}}{\sin^{4n} \cdot 2z}$

So, $(xyz)^{\frac{1}{3}} = 2 \cdot 2^{3^r}$ possible only when $\sin^{2k} 2x = 1$, $\sin^{3n} 2y = 1$, $\sin^{4m} 2z = 1$
So, required value is 3

45. 2

Sol. Use AM ≥ GM equality holds when variable are equal

So, each root = $\frac{1}{2}$

46. 2

Sol. Let x = y = 0 $\Rightarrow$ f(0) = 1. Put x = y $\Rightarrow$ 2f(x) ≤ 2 sin x + 2 $\Rightarrow$ f(x) ≤ sin x + 1

If y = –x; 1 + f(0) ≤ f(x) + f(–x) ≤ 2

$\Rightarrow$ f(–x) = 2 – f(x) ; f(–x) + f(–x) ≤ –2 sin x + 2

$\Rightarrow$ f(–x) ≤ –sin x + 1 $\Rightarrow$ 2 – f(x) ≤ –sin x + 1 $\Rightarrow$ –f(x) ≤ –sin x – 1 $\Rightarrow$ f(x) ≥ 1 + sin x

So, f(x) = 1 + sin x

47. 2

Sol. $\sqrt[8]{ad^3g^4} \le \frac{a+3d+4g}{8} \Rightarrow \sum\sqrt[8]{ad^3g^4} \le 1+3+4 \Rightarrow \frac{\sum\sqrt[8]{ad^3g^4}}{3} \le \frac{8}{3}$

SECTION – C

48. 24.00

Sol. We have, $\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} \frac{(n+1)^k - n^k}{\binom{n}{k-1}} = \lim\limits_{n\to\infty} \frac{(k-1)!\cdot k\cdot n^{k-1} + ... + (k-1)!}{(n-k+2)(n-k+3)\cdot .....\cdot n} = \frac{k!\cdot n^{k-1} + ...}{n^{k-1} + ...} = k!$

49. 2.00

Sol. Then, $b_m = 1 + \sum\limits_{k=1}^{m} k\cdot k! = 1 + \sum\limits_{k=1}^{n} (k+1)! - \sum\limits_{k=1}^{n} k! = (n+1)!$

So, $\lim\limits_{n\to\infty}\left(\frac{b_n^2}{b_{n-1}b_{n+1}}\right)^n = \lim\limits_{n\to\infty}\left(1-\frac{1}{n+2}\right)^n = e^{-1}$

50. 1.00

51. 4.00

Sol. (Q. 50-51)

f(x) = (x – 2)(x^2 – 4x + 6)

$\therefore$ g(x) = x^2 – 4x + 6 has roots = α, β

(i) L.H.S. = $\frac{\alpha^2+\beta^2}{36} = \frac{(\alpha+\beta)^2 - 2\alpha\beta}{36} = \frac{16-12}{36} = \frac{1}{9}$

(ii) 8 < g(x) ≤ 18 $\Rightarrow$ 8 < x^2 – 4x + 6 ≤ 18

$\Rightarrow$ 6 < (x – 2)^2 ≤ 16 $\Rightarrow$ $\sqrt{6} < |x-2| \le 4$

$\therefore$ |x – 2| = 3, 4